



Physics-informed neural networks for reacting flows: Species reconstruction with finite rate chemistry from sparse and noisy velocity measurements

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ABSTRACT

We present Reactive Flow Physics-informed neural networks (RF-PINNs), an extension of PINNs tailored for reactive flow problems involving finite-rate chemistry. The idea is to use only partially available and possibly noisy information of the flow field (we use velocity data without loss of generality) to infer all other missing quantities. We demonstrate that our method accurately reconstructs velocities, pressure, temperature, density, species mass fractions, reaction rates, and heat release. To our knowledge, this is the first application of PINNs to reactive flows incorporating finite-rate chemistry mechanisms, featuring 9 species and 21 reactions for hydrogen, and 21 species and 84 reactions for methane, alongside a mixture-averaged diffusion model. The central novelty of this study lies in the reconstruction of the reactive flow fields from sparse velocity measurements with incomplete boundary conditions. Velocities and temperature are reconstructed with a relative error below 1%. This was achieved through two key novelties to the RF-PINN framework. First, we introduce a specialized output layer that embeds physical constraints into the architecture by bounding the species mass fractions while ensuring their summation to unity. Second, we develop the Elemental Mass Flow Loss, a novel physics-informed loss function that enforces global conservation of elemental mass flow across the domain, eliminating the need for inlet boundary conditions. This loss enables the reconstruction of inlet profiles with high fidelity, demonstrating its potential to derive boundary conditions from sparse experimental data. RF-PINNs offer a powerful tool for combustion diagnostics by reconstructing flow fields and boundary conditions from sparse inputs, such as PIV velocity data, especially in scenarios with limited experimental access. While we demonstrate RF-PINNs on stationary hydrogen and methane flames, our methodology is readily extendable to other chemical kinetics mechanisms, fuels and operating conditions.

1. Introduction

Reactive flows play a crucial role in energy production, propulsion systems, and chemical processing, making them indispensable for modern technology. However, they are also significant contributors to climate change, underscoring the urge of using these processes as efficiently as possible while transitioning to carbon-neutral fuels such as hydrogen. Achieving optimal efficiency requires a fundamental understanding of the interplay between fluid dynamics, thermodynamics, and chemical kinetics.

Direct Numerical Simulations (DNS) and Large Eddy Simulations (LES) are cornerstones for modeling reactive flows, providing high-fidelity predictions of turbulent phenomena and chemical reactions [1–3]. However, the reliability of numerical simulations depends heavily on accurate boundary conditions, which are often challenging to obtain in practical scenarios. In contrast, experimental methods, which rely on costly diagnostics, such as high-speed cameras and lasers, can directly

measure flow properties and provide valuable insights. Techniques like Particle Image Velocimetry (PIV) and Coherent Anti-Stokes Raman Spectroscopy provide valuable data but are limited in spatial and temporal resolution, are often noisy, and require extensive calibration. This gap between simulations and experimental data motivates the need for methods that can integrate sparse measurements into robust flow field reconstructions.

Physics-informed Neural Networks (PINNs) [4] have recently emerged as a powerful tool for solving partial differential equations in diverse fields, including fluid mechanics [5]. Combining the data-driven approach with embedded physical laws enables PINNs to generalize well with minimal data, making them particularly effective when experimental measurements are incomplete or sparse [6]. While PINNs have shown promise in fluid flow modeling, their application to reactive flows remains underexplored [7,8]. Existing studies have successfully applied PINNs to laminar [9], turbulent [10], and sooting

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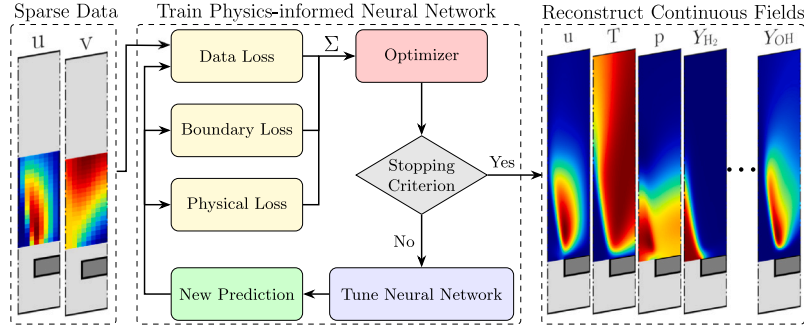


Fig. 1. Illustration of the RF-PINN framework: (Left) Sparse synthetic PIV data partially covering the domain, leaving large regions to be inferred by our method. (Middle) Training workflow featuring a specialized loss function that integrates data loss (measurement adherence), boundary loss (boundary condition compliance), and physical loss (governing equation fidelity). (Right) Fully reconstructed flow fields across the entire domain, including velocity, temperature, pressure, and species.

flames [11], focusing primarily on simplified diffusion models and global chemical kinetics mechanisms with one or two reactions. While computationally efficient, global mechanisms present a strong simplification for the underlying chemical kinetics, rendering them insufficient for reconstructing species distributions with high fidelity. The detailed reconstruction of species concentrations or reaction rates in reactive flows has not been demonstrated, despite the critical importance of these quantities in understanding and evaluating combustion systems.

In this work, we present an extension to RF-PINNs [12], a physics-informed framework that enables the reconstruction of detailed reacting flow fields using only sparse velocity data and operating conditions. By incorporating complex chemical mechanisms and thermophysical models, RF-PINNs accurately reconstruct velocity, temperature, pressure, species concentrations, density, material properties, reaction rates, and heat release rate. (Fig. 1)

To overcome the challenges of training PINNs for reactive flows, we introduce two novelties that enhance both the accuracy and robustness of our framework. The first innovation is the *Bounded Output Layer*, a tailored architectural adjustment that ensures species mass fractions remain physically bounded and sum to unity. By embedding these physical constraints directly into the network, this modification significantly enhances robustness and accelerates convergence during training. The second is the *Elemental Mass Flow Loss*, a novel physics-informed loss function designed to enforce the global conservation of elemental mass flow across the domain. This approach removes the need for detailed inlet boundary conditions by relying solely on global parameters, such as the equivalence ratio and total mass flow rate. Notably, this loss function not only eliminates dependence on precise inlet profiles but also enables their reconstruction from sparse experimental data.

In contrast to other data-driven studies, RF-PINNs require minimal training, relying instead on the physics-informed loss function for accurate reconstructions. Each model is trained for a specific configuration and therefore does not generalize across cases in the traditional machine learning sense. However, similar to a Computational Fluid Dynamics (CFD) solver, the framework itself is generalizable as the same physical loss formulation and training procedure can be applied to different operating conditions, flow regimes, or chemical mechanisms without modification. Adapting to a new configuration simply requires training a new model, analogous to running a new CFD case. By jointly minimizing multiple loss terms, RF-PINNs enforce physics, incorporate measurements, and embed integral constraints to reconstruct flow fields and unknown boundary in an inverse manner. This approach eliminates the need for costly data assimilation or optimization procedures of conventional CFD solvers.

We demonstrate the framework for stationary premixed hydrogen and methane flames governed by complex mechanisms – 9 species and 21 reactions for hydrogen, and 21 species and 84 reactions for methane

– alongside a mixture-averaged diffusion model to capture the high diffusivity of hydrogen. The framework yields accurate predictions of all key flow variables, paving the way for leveraging experimental data to enhance the understanding and modeling of combustion processes.

The paper is structured as follows: Section 2 describes the data generation process using a DNS solver, which provides the ground truth for model evaluation. Section 3 introduces the RF-PINN framework, including losses, architecture, and training. Section 4 shows the reconstruction results for both clean and noisy velocity data. The supplementary material includes the full set of modeling equations, the derivation of the Elemental Mass Flow Loss, and additional results.

2. Data generation: DNS of H_2 and CH_4 flames

We study a laminar slit flame configuration [13]. It consists of eight narrow rectangular slit burners to create laminar, premixed flames that extend in one direction. CH_4 and air are premixed at an equivalence ratio (ER) of $\phi = 0.8$ and enter the plenum at 293 K with a bulk velocity U_{in} of 0.4 m s^{-1} . This setup has been reproduced numerically in different studies [14–16]. In this work, we simulate the reference CH_4 case and we introduce an additional case by replacing CH_4 with H_2 . The numerical setup and boundary conditions are illustrated in Fig. 2. The DNS domain extends upstream to include the plenum and the contraction, whereas the PINN is trained on a truncated domain that begins inside the slit (green outline), omitting the upstream geometry.

2.1. DNS code *dmfFoam*

Both numerical simulations are conducted with a customized OpenFOAM [17] solver developed in-house. The DNS solver *dmfFoam* links the chemical kinetics solver Cantera [18] with the base OpenFOAM solver *reactingFoam*. Therefore, it combines OpenFOAM’s capabilities to solve the compressible Navier–Stokes, species transport and energy equations together with Cantera’s routines for the computation of detailed thermo-physical and transport properties. The governing equations implemented in *dmfFoam* are the conservation of mass, momentum, species, and energy [1]:

$$\frac{\partial \rho}{\partial t} + \frac{\partial \rho u_j}{\partial x_j} = 0, \quad (1)$$

$$\frac{\partial \rho u_i}{\partial t} + \frac{\partial \rho u_i u_j}{\partial x_j} = -\frac{\partial p}{\partial x_i} + \frac{\partial \tau_{ij}}{\partial x_j}, \quad (2)$$

$$\frac{\partial \rho Y_k}{\partial t} + \frac{\partial \rho u_i Y_k}{\partial x_i} = -\frac{\partial J_{k,i}}{\partial x_i} + \dot{\omega}_k, \quad (3)$$

$$\frac{\partial \rho H}{\partial t} + \frac{\partial}{\partial x_i} (\rho u_i H) = \frac{\partial p}{\partial t} - \frac{\partial q_i}{\partial x_i} + \dot{\omega}_T, \quad (4)$$

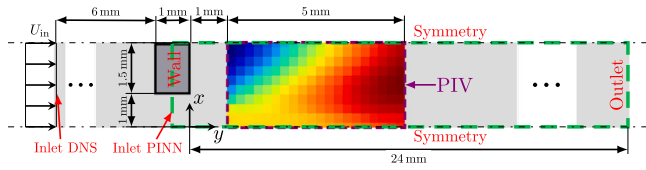


Fig. 2. Sketch of the slit flame configuration. The DNS includes the upstream region with a uniform inflow profile. The green frame indicates the smaller computational domain used for RF-PINNs, starting at the slit center where the inlet is no longer uniform. Red labels show the respective boundaries. The purple frame marks the synthetic PIV window used as velocity data.

where u , ρ , Y_k , $\dot{\omega}_k$, p , H , $\dot{\omega}_T$ are respectively the velocity vector, density, species k mass fraction, species k chemical source term, pressure, total enthalpy and heat release rate. The viscous stress tensor τ , the species k diffusive flux J_k and the energy flux q are:

$$\tau_{ij} = -\frac{2}{3}\mu\frac{\partial u_k}{\partial x_k}\delta_{ij} + \mu\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right), \quad (5)$$

$$J_{k,i} = -\rho Y_k (V_{k,i} + V_i^c), \quad (6)$$

$$q_i = -\lambda\frac{\partial T}{\partial x_i} + \sum_{k=1}^N h_{s,k} J_{k,i}, \quad (7)$$

where the dynamic viscosity μ is computed from Wilke mixture rule [19]. The species diffusion velocities V_k are computed according to Hirschfelder and Curtis approximation [20] and V^c is the correction velocity that ensures global mass conservation. λ is the thermal conductivity, and T the temperature. Transport properties are calculated with the mixture-averaged diffusion model; additional material property details are given in the supplementary material.

2.2. Reaction mechanisms

Combustion of H_2 /air and CH_4 /air mixtures is modeled respectively using the San Diego mechanism [21] and the DRM19 mechanism [22], which comprise 9 species/21 reactions and 21 species/84 reactions, respectively. We calculated the one-dimensional laminar flame speed, s_L , using Cantera [18] for the reference case operating conditions with the DRM19 mechanism, yielding a value of 0.294 m s^{-1} . The ER for the H_2 /air case was set to 0.43 to match this flame speed.

2.3. DNS code validation

We validate our code using one-dimensional freely propagating flames. Figure S1 of the supplementary material compares the one-dimensional flame structures of Cantera and *dmfFoam* for the two chemical kinetics mechanisms. The results show excellent agreement, with maximum deviations of less than 2%. Table S1 in the supplementary material compares the flame speed s_L , flame thickness δ_f and adiabatic temperature T_{ad} . The values show excellent agreement, with the largest discrepancy being 2% for the flame thickness when using the San Diego mechanism.

2.4. Numerical setup of the laminar slit flame

The geometry of the experimental setup (Fig. 2), enables a two-dimensional numerical analysis by modeling only half of a single flame. A symmetry boundary condition is applied on the sides of the domain, with a no-slip boundary condition imposed on the slit wall. At the outlet, a non-reflective boundary condition is applied. We used the second order implicit time scheme backward Euler and a fourth order cubic scheme for spatial discretization. The mesh is created with a local resolution of $12.5 \mu\text{m}$ in order to resolve the flame front with

Table 1

Operating conditions for the two dimensional laminar slit flame Cases. Case 2 corresponds to the reference setup of Kornilov et al. [13]:

Case	Fuel	ϕ	Mechanism	T_{in} [K]	T_{wall} [K]	U_{in} [$\frac{m}{s}$]
1	H_2	0.43	San Diego [21]	293	373	1
2	CH_4	0.8	DRM19 [22]	293	373	0.4

at least 10 grid points. According to previous studies [13,15,23] the wall temperature is set to a constant value of 373 K. The operating conditions for the two cases are listed in Table 1. In Case 1, CH_4 is replaced with H_2 , and the inlet velocity U_{in} is increased from 0.4 to 1 m s^{-1} .

2.5. DNS results

In subsequent sections, the DNS results serve as the reference for comparison with the PINN predictions. Additional figures of the results are included in the supplementary material. For Case 2, the DNS flame length aligns well with the experimental data. For Case 1, which lacks experimental measurements, the flame length is consistent with the findings of [16].

3. RF-PINN framework

We develop RF-PINNs as physics-driven framework for reconstructing continuous reactive flow fields from minimal data.

3.1. Framework overview

The RF-PINN approximates the solution $\hat{u}(x, y)$ of the reacting flow, with prediction $u_\theta(x, y)$ parameterized by the trainable weights and biases summarized as θ . The network takes the spatial coordinates x and y as inputs and predicts a total of $4 + N$ output quantities: the velocity components, temperature, pressure, and the mass fractions of N chemical species. Based on the predicted quantities, we calculate material properties including density, isobaric heat capacity, enthalpy, entropy, thermal conductivity, viscosity, and mixture-averaged diffusion coefficients, using the same models employed in the simulations. The equations used to compute the material properties are detailed in the supplementary material. Reaction and heat release rates are computed using the respective chemistry mechanisms analogous to Cantera [18]. The neural network architecture enables the computation of spatial derivatives of both the predicted quantities and the material properties derived from them through automatic differentiation. These form the basis for the physics-based loss functions described below.

3.2. Architecture

RF-PINNs employ a feedforward neural network with hyperbolic tangent activation functions in the hidden layers. The network consists of 5 layers, with 54 neurons per layer for Case 1 and 64 neurons per layer for Case 2. The output layer is tailored to the specific flow quantities: it uses linear activation for velocities and pressure, while temperature predictions are bounded using a scaled offset sigmoid activation function. It ensures that temperature predictions remain within $T \in [293 \text{ K}, 2100 \text{ K}]$. For species mass fractions, we introduce a modified output layer. First, a scaled and offset sigmoid function σ is applied to the unbounded species prediction $Y_k^* \in \mathbb{R}$:

$$\hat{Y}_k = (\hat{Y}_{k,max} - \hat{Y}_{k,min}) \sigma(Y_k^*) + \hat{Y}_{k,min}, \quad (8)$$

ensuring $\hat{Y}_k \in [\hat{Y}_{k,min}, \hat{Y}_{k,max}]$. The predictions are then normalized:

$$Y_k = \frac{\hat{Y}_k}{\sum_{i=1}^N \hat{Y}_i}. \quad (9)$$

While this normalization step relaxes the initial bounds introduced in Eq. (8), it guarantees that the predicted mass fractions satisfy the constraint $\sum_k Y_k = 1$. For most species, the lower bounds are set to zero, with exceptions for N_2 and O_2 . As both flames operate in the lean regime we apply non-zero lower bounds to N_2 and O_2 . Given that N_2 represents the largest fraction of the mixture, the bounded prediction helps to establish a good approximation of the material properties early during the training. Upper bounds for the fuel and oxygen mass fractions are constrained to their respective inlet values and the upper bounds for other species are estimated based on the results from Cantera simulations. Overly restrictive bounds can prevent the network from capturing the correct solution, whereas looser bounds still allow for stable training. Excessively loose bounds, exceeding expected values by several orders of magnitude, cause the sigmoid activation to produce vanishing gradients and slow down convergence. This species layer improves convergence speed and robustness while adding less than 1% to total training cost.

3.3. Loss functions

RF-PINNs are trained using a specialized training loss $\mathcal{L}_{\text{train}}(\Theta)$ that is a weighted sum of data-driven and physical constraints:

$$\mathcal{L}_{\text{train}}(\Theta) = \Lambda_{\text{data}} \mathcal{L}_{\text{data}}(\Theta) + \Lambda_{\text{bc}} \mathcal{L}_{\text{bc}}(\Theta) + \Lambda_{\text{physics}} \mathcal{L}_{\text{physics}}(\Theta) + \Lambda_{\text{emf}} \mathcal{L}_{\text{emf}}(\Theta). \quad (10)$$

It can be interpreted as a scalarized multi-objective optimization problem of competing objectives such as boundary conditions or momentum conservation. We manually tune the loss weights Λ to balance the objectives. While automated tuning methods have shown promise for single-PDE systems [24], extending them to multi-PDE systems remains an open challenge.

The **data loss** promotes adherence to measurements in the domain interior. It is evaluated over N_{data} data points $\{x_{\text{data}}^1, \dots, x_{\text{data}}^{N_{\text{data}}}\}$ and is defined as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{j=1}^{N_{\text{data}}} \left\| \mathbf{u}_{\Theta}(x_{\text{data}}^j) - \hat{\mathbf{u}}(x_{\text{data}}^j) \right\|^2. \quad (11)$$

The **boundary loss** promotes the adherence to Dirichlet and Neumann boundary conditions. It is evaluated over N_{bc} boundary points $\{x_{\text{bc}}^1, \dots, x_{\text{bc}}^{N_{\text{bc}}}\}$:

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathbf{u}_{\Theta}(x_{\text{bc}}^j) - \hat{\mathbf{u}}(x_{\text{bc}}^j) \right\|^2 + \left\| \frac{\partial \mathbf{u}_{\Theta}(x_{\text{bc}}^j)}{\partial \mathbf{n}(x_{\text{bc}}^j)} - \frac{\partial \hat{\mathbf{u}}(x_{\text{bc}}^j)}{\partial \mathbf{n}(x_{\text{bc}}^j)} \right\|^2, \quad (12)$$

where $\mathbf{n}$ denotes the surface normal vector.

The **physical loss** promotes adherence of the predictions to the governing equations for continuity, momentum, species transport, and energy. For training, the stationary form of the governing Eqs. (1)–(4) are reformulated in a nondimensionalized form by scaling all variables with appropriate reference quantities. [12] The residuals $\mathcal{R}_{\text{mass}}[u_{\Theta}]$, $\mathcal{R}_{\text{momentum}}[u_{\Theta}]$, $\mathcal{R}_{\text{species}}[u_{\Theta}]$ and $\mathcal{R}_{\text{energy}}[u_{\Theta}]$ quantify the deviation between the left- and right-hand sides of the governing equations for a given prediction. For training, the mean squared error (MSE) of each residual is computed on N_c collocation points $\{x_c^1, \dots, x_c^{N_c}\}$, forming the individual loss terms:

$$\mathcal{L}_q = \frac{1}{N_c} \sum_{i=1}^{N_c} (\mathcal{R}_q[\mathbf{u}_{\Theta}](x_c^i))^2, \quad (13)$$

where $q \in \{\text{mass, momentum, species, energy}\}$. The total physical loss is a weighted sum of these terms, where Λ_q are the loss weights of the individual terms:

$$\mathcal{L}_{\text{physics}} = \sum \Lambda_q \mathcal{L}_q. \quad (14)$$

Table 2

PINN boundary conditions in SI units. φ represents v, T and Y_k .

Boundary	Dirichlet	Neumann
Inlet	$Y_k = Y_{k,\text{in}}$	–
Wall	$u = v = 0, T = 373$	$\frac{\partial p}{\partial n} = 0, \frac{\partial Y_k}{\partial n} = 0$
Symmetry	$u = 0$	$\frac{\partial \varphi}{\partial n} = 0$
Outlet	$p = 101,325$	$\frac{\partial u}{\partial n} = 0, \frac{\partial \varphi}{\partial n} = 0$

3.3.1. Elemental mass flow loss

When training PINNs for reactive flows, we observed that the inlet mass fractions are quickly learned due to imposed boundary loss. However, the composition of the downstream prediction is governed through the species transport equations, which are gradually learned throughout the training. Errors in composition accumulate throughout the domain, resulting in predicted mixture compositions incompatible with the operating conditions. To address this, we propose a novel physics-informed loss function: the **Elemental Mass Flow (EMF) Loss**. The loss penalizes discrepancies between the inlet elemental mass flow rate $\dot{m}_{e,\text{in}}$, computed from the equivalence ratio and the bulk mass flow, and the predicted elemental mass flow rate $\dot{m}_e(y_s)$, for each element present in the flow $e \in \{\text{H, C, N, } \dots, \text{O}\} = \mathcal{E}$. By enforcing elemental conservation across the domain, this global constraint enhances both prediction accuracy and convergence stability, particularly in cases where the inlet profile is not given. A detailed derivation of the loss function is provided in the supplementary material. In this work, the loss is evaluated at N_s linearly spaced sections $y_s \in [-0.5, 24]$ mm in the domain in the y -direction where the bulk elemental mass flow is integrated across the x -axis:

$$\mathcal{L}_{\text{emf}} = \frac{1}{N_s} \sum_{e \in \mathcal{E}} \sum_{s=1}^{N_s} (\dot{m}_{e,\text{in}} - \dot{m}_e(y_s))^2 \quad (15)$$

The computational overhead of this loss depends on the number of sections and integration points and is typically below 10% of the total training cost.

3.4. Boundary conditions & training data

The computational domain is depicted in Fig. 2. The PINNs are trained to reconstruct the flow fields downstream of the slit. Due to the truncation, the non-uniform velocity and temperature profiles at the new inlet are unknown. These boundary conditions are left **unconstrained**. This configuration would be ill-posed for conventional CFD solvers, which require fully prescribed inlet conditions. Instead we use RF-PINNs to recover the profiles from the governing equations, downstream velocity measurements, ER and mass flow information, effectively solving the problem in an inverse manner. The full set of used boundary conditions is summarized in Table 2.

To emulate PIV data, a rectangular window (Fig. 2) is extracted from the **simulation velocity fields only**, leaving a 1 mm gap between the slit and window, as PIV measurements are typically hard to obtain in the near wall region. Although the DNS mesh contains approximately 90,000 cells in this region, the fields are downsampled to 200 points by interpolating the original fields from the respective DNS onto a uniform grid (see Fig. 2). This coarse sampling is intended to mimic coarse experimental PIV data, that is also subjected to noise (Section 4.1). These interpolated velocities are then enforced through the data loss term. Temperature, pressure, and species fields inside the domain are not used as training data and are instead reconstructed implicitly through the physical losses.

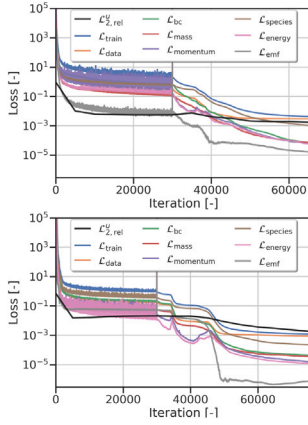


Fig. 3. Loss history for Case 1 (top) and Case 2 (bottom)

3.5. Training & accuracy of RF-PINNs

Both cases are initially trained with 30,000 iterations with the SOAP optimizer [25]. Afterwards, training continues with SS-Broyden [26] until convergence. Both cases use 200 data points, 3,000 boundary points, and 60,000 collocation points generated via uniform Hammerley sampling, with full batch updates. The Elemental Mass Flow Loss is computed across 100 sections with 300 integration points each. On a NVIDIA A100 GPU, training Case 1 takes 4 h, while the increased complexity of Case 2's chemical kinetic mechanism along with the larger neural network extends the training time to 6 h.

Since the training loss does not directly reflect prediction accuracy, we compute the relative $\mathcal{L}_2$ error for each predicted quantity ψ , computed on a uniform grid, and only evaluated every 5000 iterations to reduce cost. This error is used solely for evaluation and is not included in the training loss:

$$\mathcal{L}_{2,\text{rel}}^\psi(\Theta) = \frac{\|\psi - \psi_{\text{DNS}}\|_2}{\|\psi_{\text{DNS}}\|_2}. \quad (16)$$

4. Results: Quality of reconstruction

Fig. 3 presents the training losses for both cases, showing an accelerated convergence after switching to SS-Broyden. Both cases show very similar convergence behavior. During later stages of training, the data and species losses are the largest contributions to the training loss. Notably, the network ceases to further fit the velocity data, instead prioritizing minimization of the physical and boundary losses. This behavior highlights the benefit of dual supervision: rather than overfitting to the data, the physical constraints guide the network toward more physically consistent solutions.

Figs. 4 and 5 compare the DNS reference (left) with the RF-PINN reconstructions from PIV data (right) for Cases 1 and 2 using both BOL and EMF. Across both cases, the reconstructions show excellent agreement in velocity, temperature, pressure, heat release rate, and species mass fractions. The predictions show that RF-PINNs are able to infer the thermophysical state from the velocity data in the area of the measurements, but also up- and downstream of the measurements. For Case 1 the largest deviations occur in the recirculation zone downstream of the wall, and are most notable for minor species such as H_2O_2 . We suspect that this is due to the lack of direct measurement data in this region and the high sensitivity of minor species to boundary conditions. Case 2 shows strong agreement for all reconstructions, including all minor species. Some minor species mass fraction predictions (such as HCO) tend to be larger than the DNS reference and show a slightly thickened reaction zone. Profiles along the flame surface normal at mid-flame length are shown in Fig. 6, confirming the good agreement in flame

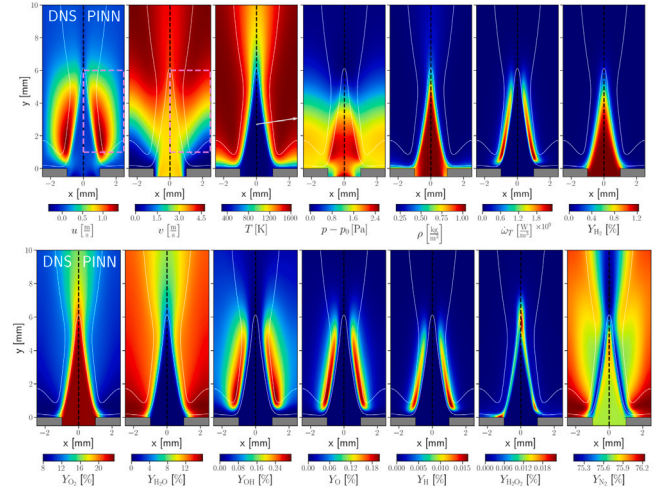


Fig. 4. Flow fields of the H_2 flame (Case 1) with the San Diego mechanism [21]. Left: DNS field. Right: RF-PINN reconstruction using the BOL and EMF loss. White isolines mark 750 K and 1500 K, with the purple frame indicating the PIV window.

structure. Quantitatively, the $\mathcal{L}_{2,\text{rel}}$ errors of primary flow quantities are summarized in Table 3 for Cases 1 and 2. The results highlight the limitations of the baseline PINN, which employs a sigmoid output layer to constrain species mass fractions to $[0, 1]$. Because intermediate species often contribute less than 10^{-4} of the total mass fraction, applying loose bounds to their outputs causes the sigmoid function to saturate, producing vanishing gradients that impede efficient training and lead to reconstruction errors exceeding 100%. These limitations are effectively mitigated by the BOL, which enables accurate predictions. All subsequent tests are performed using BOL. The EMF loss has a minor effect in Case 1, but substantially improves accuracy in Case 2, likely due to the additional carbon mass flow constraint. This suggests that EMF becomes increasingly beneficial as the number of elements in the mechanism grows. Overall, the combination of BOL and EMF enable RF-PINNs to reconstruct velocities and temperatures with less than 1% relative error and while major species (peak mass fraction $> 1\%$) fall below 4%, despite unconstrained inlet velocity and temperature profiles. The computational overhead per training iteration is less than 1% for BOL and around 7%–8% for EMF. Fig. 7 shows the reconstructed profiles along the centerline ($x = 0$ mm) and at the dump ($y = 0$ mm) (see Fig. 2 for the location). The profiles match the DNS reference closely, confirming the network's ability to infer consistent solutions up- and downstream of the PIV window, showing the potential of the approach in inferring boundary conditions. Accurate upstream predictions require sufficient constraints, such as velocity in our case, to ensure the RF-PINN produces reliable results. Without adequate constraints, the problem becomes underdetermined and allows for multiple plausible upstream profiles that satisfy the downstream constraints. Nevertheless, the reconstruction quality remains high, with an error of less than 2%, which likely falls within the range of experimental uncertainties.

4.1. Robustness of predictions to noise

To assess the robustness of the proposed framework the reconstruction accuracy was evaluated with synthetic noise added at each data point. For every point and velocity component, noise was drawn from a Gaussian distribution:

$$\hat{u}_{\text{noisy}} = \hat{u} + \sigma \eta, \quad \eta \sim \mathcal{N}(0, 1). \quad (17)$$

We experimented with standard deviations σ ranging from 0.1 m/s to 0.5 m/s. For consistency, a fixed noise realization η was scaled across runs.

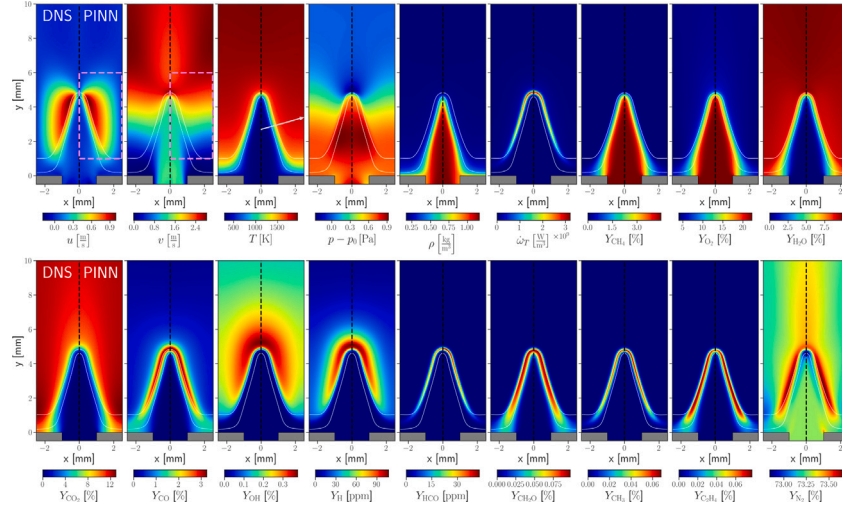


Fig. 5. Flow fields of the CH_4 flame (Case 2) with the DRM19 Mechanism [22]. Left: DNS field. Right: RF-PINN reconstruction using the BOL and EMF loss. White isolines mark 750 K and 1500 K, with the purple frame indicating the PIV window.

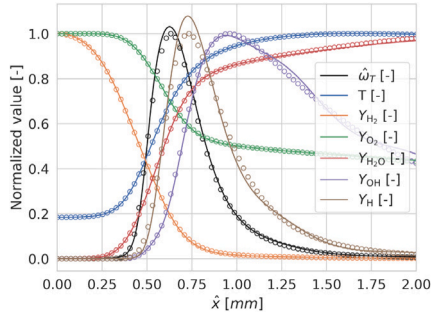


Fig. 6. Comparison of normalized RF-PINN (lines) and DNS (circles) profiles for Case 1 along the flame-normal (see gray arrow in Fig. 4 for position); Case 2 profiles are provided in the supplementary material.

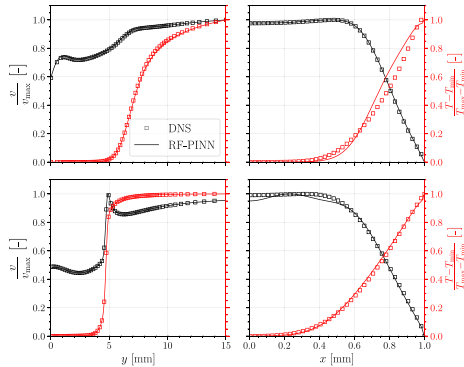


Fig. 7. Comparison of RF-PINN reconstructed profiles for Case 1 (top) and Case 2 (bottom). Left: Centerline ($x = 0$ mm). Right: Inlet ($y = 0$ mm).

Table 4 summarizes the reconstruction errors for Case 1 with increasing noise amplitudes, comparing results with and without the EMF loss. Errors increase monotonically with noise amplitude, but the EMF loss consistently improves the reconstruction accuracy, demonstrating that the integral constraint enhances robustness to noise. Even at standard deviations of $\sigma = 0.5$ m/s velocity and temperature errors remain below 10% and major species stay within 18%. The reconstruction of intermediate species such as OH is more sensitive to noise, with errors exceeding 50% at high noise levels. These results indicate that the

Table 3

Reconstruction errors $\mathcal{L}_{2,\text{rel}}^\psi$ for Case 1 & 2 for velocity, temperature, and selected mass fractions, comparing the Baseline, BOL, and BOL & EMF approaches.

Case 1	v	T	Y_{H_2}	Y_{O_2}	$Y_{\text{H}_2\text{O}}$	Y_{OH}
Baseline	7.3%	54.3%	977%	60.9%	100%	100%
BOL	0.2%	0.8%	5.4%	1.6%	1.9%	9.1%
BOL & EMF	0.2%	0.6%	3.1%	1.7%	2.3%	9.1%
Case 2	v	T	Y_{CH_4}	Y_{O_2}	$Y_{\text{H}_2\text{O}}$	Y_{CO}
Baseline	234%	40.6%	100%	1034%	102%	1310%
BOL	0.5%	3.9%	9.7%	6.4%	8.3%	91.7%
BOL & EMF	0.2%	0.1%	0.9%	1.3%	0.7%	3.4%

Table 4

Reconstruction errors $\mathcal{L}_{2,\text{rel}}^\psi$ on Case 1 for different noise levels σ without and with the EMF loss.

Without EMF loss						
σ [m/s]	v	T	Y_{H_2}	Y_{O_2}	$Y_{\text{H}_2\text{O}}$	Y_{OH}
0.1	1.1%	2.9%	5.4%	5.2%	5.9%	26.2%
0.25	2.7%	19.1%	89.3%	29.0%	36.2%	75.2%
0.5	3.8%	11.2%	25.4%	17.2%	19.5%	58.0%
With EMF loss						
σ [m/s]	v	T	Y_{H_2}	Y_{O_2}	$Y_{\text{H}_2\text{O}}$	Y_{OH}
0.1	1.2%	2.7%	3.8%	5.1%	5.6%	26.4%
0.25	2.1%	4.6%	5.2%	7.4%	8.2%	39.4%
0.5	3.9%	8.9%	17.5%	12.5%	13.8%	54.7%

framework is robust to realistic noise levels, although further work is needed to improve accuracy under severe noise conditions.

The Supplementary Material includes the reconstructed fields for noisy data in both cases and the reconstruction error table for Case 2. It also presents further analyses on the impact of biased noise, collocation point sensitivity, and data resolution.

5. Conclusion

We have demonstrated the capability of the RF-PINN framework to reconstruct reacting flow fields from sparse velocity measurements. The results show that the framework achieves high fidelity predictions across velocity, pressure, temperature, and species mass fractions, accounting for complex chemical mechanisms and mixture-averaged diffusion. The Bounded Output Layer accelerates convergence during

training by enforcing bounds and ensuring that species mass fractions sum to unity. The novel Elemental Mass Flow Loss ensures consistent convergence without requiring explicit inlet boundary conditions. Notably, the framework recovers inlet profiles with minor deviations, highlighting its potential to derive boundary conditions from experimental data. This capability could significantly enhance numerical simulations, as their accuracy depends on the supplied boundary conditions. When subjected to noisy measurements, the reconstruction accuracy decreases as expected but remains robust overall. Intermediate species exhibit greater sensitivity to noise, suggesting that incorporating complementary measurements such as temperature, pressure, or species concentrations can further improve reconstruction accuracy, particularly in high-noise scenarios. Overall, the study demonstrates that RF-PINNs provide a flexible and physically consistent framework for reconstructing reacting flows from limited and imperfect data, bridging the gap between experimental measurements and numerical modeling.

CRedit authorship contribution statement

Fabio Froberg: Writing – original draft, Visualization, Software, Methodology, Investigation, Data curation. **Pablo Kandel:** Writing – original draft, Visualization, Software, Data curation. **Abdulla Ghani:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Novelty and significance statement

- This work presents the first use of PINNs to reconstruct reactive flows with complex chemical kinetics from minimal data.
- Unlike other machine learning approaches that require large datasets, RF-PINNs achieve accurate reconstructions with as few as 200 velocity measurements and no pretraining. Key innovations include a custom output layer that bounds species mass fractions while enforcing their sum to unity, the incorporation of integral elemental mass flow conservation as a constraint, and a mixture-averaged diffusion model to capture diffusive species, enabling robust reconstruction of all flow fields from sparse and noisy data.
- Its significance stems from its ability to reconstruct inaccessible flow quantities without requiring all boundary conditions.
- While this study uses sparse velocity data, the framework can accommodate diverse inputs (e.g., pressure, temperature, or species). This enables deeper insight into the thermochemical state of reacting flows and accurate reconstruction of boundary conditions, bridging the gap between experiments and high-fidelity simulations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2025.105957>.

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